

DETECTION RULE DOCUMENTATION

Registry Run Key Persistence — Behavioral Detection

Rule ID: DET-001 **Rule Name:** Suspicious Registry Run Key Modification via PowerShell
Author: Olayinka Oyetade **Created:** May 5, 2026 **MITRE ATT&CK:** T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys **Severity:** HIGH **Status:** ACTIVE

1. RULE OVERVIEW

Purpose

This detection rule identifies unauthorized modifications to Windows Registry Run keys — a common persistence mechanism used by malware to survive reboots. The rule specifically focuses on modifications originating from PowerShell, cmd.exe, or other scripting engines, which are strong indicators of malicious or unauthorized activity.

What This Rule Detects

- PowerShell writing entries to HKCU or HKLM Run keys
- Any process writing to autorun registry locations using unusual image paths
- Masqueraded registry key names (e.g., "WindowsUpdate", "SystemService") pointing to non-standard executable paths

Why Behavioral Detection Works Better Than Signatures

Signature-based detection (e.g., antivirus) looks for known malware file hashes. Attackers easily bypass this by modifying the malware binary. This rule detects the **behavior** — the act of writing to a Run key — regardless of what the payload is or what it's called. The behavior is consistent across Emotet, TrickBot, Remcos, and thousands of other malware families.

2. DATA REQUIREMENTS

Requirement	Value
SIEM	Splunk Enterprise
Data Source	Sysmon for Windows
Sysmon Version	v14+ (with XML event format)
Sysmon Config	Must include Event ID 13 (RegistryEvent - Value Set)
Splunk Add-on	Splunk Add-on for Microsoft Sysmon
Index	endpoint
Sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

Required Sysmon Configuration

The following must be present in your Sysmon configuration (sysmonconfig.xml):

```
<RegistryEvent onmatch="include">
  <TargetObject condition="contains">CurrentVersion\Run</TargetObject>
  <TargetObject condition="contains">CurrentVersion\RunOnce</TargetObject>
</RegistryEvent>
```

3. DETECTION QUERY

Primary Query (High Fidelity)

```
index=endpoint EventCode=13 "powershell.exe" "CurrentVersion\\Run"
| table _time, host, User, Image, TargetObject, Details
| sort -_time
```

Explanation

Component	Purpose
index=endpoint	Scopes search to the endpoint security index
EventCode=13	Filters for Sysmon Registry Value Set events only

"powershell.exe"	Limits results to modifications made by PowerShell (high-risk source)
"CurrentVersion\\Run"	Targets the specific registry path used for persistence
table	Formats results for analyst readability
sort -_time	Shows most recent events first

Broader Query (Full Coverage, Higher False Positive Rate)

```
index=endpoint EventCode=13
(TargetObject="*CurrentVersion\\Run*" OR TargetObject="*CurrentVersion\\RunOnce*"
| stats count by _time, host, User, Image, TargetObject, Details
| sort -_time
```

Use this query in production to catch modifications from all processes, not just PowerShell. Requires tuning to whitelist legitimate software.

Enriched Query (Production-Ready)

```
index=endpoint EventCode=13
(TargetObject="*CurrentVersion\\Run*" OR TargetObject="*CurrentVersion\\RunOnce*"
| where NOT (Image="*msiexec.exe" OR Image="*setup.exe" OR Image="*installer.exe"
| eval risk_score=if(match(Image,"powershell|cmd|wscript|cscript"),80,40)
| table _time, host, User, Image, TargetObject, Details, risk_score
| sort -risk_score, -_time
```

4. ALERT CONFIGURATION

Setting	Value
Alert Name	ALERT: Suspicious Registry Run Key Modification
Search Schedule	Every 5 minutes (real-time monitoring)
Trigger Condition	Number of results > 0
Trigger Mode	Once per result
Action	Email notification to SOC analyst

Email Subject	[HIGH] Registry Persistence Detected — {{host}}
Throttle	1 alert per host per 60 minutes

Email Alert Template

ALERT: Suspicious Registry Run Key Modification Detected

Severity: HIGH
Host: {{host}}
Time: {{_time}}
User: {{User}}
Process: {{Image}}
Registry Key: {{TargetObject}}
Value Written: {{Details}}

Investigate immediately.
MITRE ATT&CK: T1547.001 – Boot or Logon Autostart Execution

- Analyst Action Required:
- 1. Verify if this modification is authorized
 - 2. Check if malware.exe or the referenced binary exists on disk
 - 3. Review process tree for parent-child relationship
 - 4. Check for additional persistence mechanisms

5. TUNING GUIDANCE

Known False Positives

Process	Reason	Action
msiexec.exe	Software installers legitimately write Run keys	Whitelist
setup.exe	Third-party installer pattern	Whitelist with review
OneDrive.exe	Microsoft OneDrive adds itself to Run keys	Whitelist
Steam.exe	Gaming client adds startup entry	Whitelist
Teams.exe	Microsoft Teams auto-start	Whitelist

Reducing False Positives

Add exclusions directly to the SPL query:

```
| where NOT (Image LIKE "%msiexec.exe"  
            OR Image LIKE "%OneDrive.exe"  
            OR Image LIKE "%Teams.exe")
```

Increasing Fidelity

Combine this rule with additional signals to reduce noise and increase confidence:

- EventCode 1 (Process Create) — What process launched PowerShell?
- EventCode 11 (File Create) — Was a new file dropped in the same timeframe?
- EventCode 3 (Network Connect) — Is the process making outbound connections?

A Run key modification from PowerShell that was itself launched by an Office macro (winword.exe → powershell.exe → registry write) is a near-certain malicious chain.

6. INVESTIGATION PLAYBOOK

When this alert fires, follow these steps:

Step 1 — Verify the alert

```
index=endpoint EventCode=13 host=<affected_host>  
| table _time, User, Image, TargetObject, Details
```

Step 2 — Check the parent process (what launched PowerShell?)

```
index=endpoint EventCode=1 host=<affected_host> Image="*powershell*"  
| table _time, ParentImage, ParentCommandLine, CommandLine
```

Step 3 — Check if the payload file exists

```
index=endpoint EventCode=11 host=<affected_host>  
| table _time, Image, TargetFilename
```

Step 4 — Check for network activity

```
index=endpoint EventCode=3 host=<affected_host>  
| table _time, Image, DestinationIp, DestinationPort
```

Step 5 — Determine scope

```
index=endpoint EventCode=13 TargetObject="*CurrentVersion\\Run*"
| stats count by host
| sort -count
```

Are other hosts affected? If yes, escalate immediately.

7. RULE PERFORMANCE

Metric	Value
Time to Detect	< 5 minutes from attack to alert
False Positive Rate (Lab)	Low — PowerShell + Run key filter is very specific
Detection Coverage	Catches technique regardless of payload name or hash
MITRE Coverage	T1547.001 (primary), adjacent to T1059.001 (PowerShell)

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documentation for portfolio and interview use